MILITARY SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, FLAT CABLE, NONENVIRONMENTAL, PLUG WITH DUAL POLARIZATION, ROUND CONDUCTOR, INSULATION DISPLACING, NON-REMOVABLE SOCKET CONTACTS (.100 X .100 SPACING)

This specification sheet is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the flat cable connectors described herein shall consist of this specification sheet and the latest issue of MIL-C-83503(USAF).

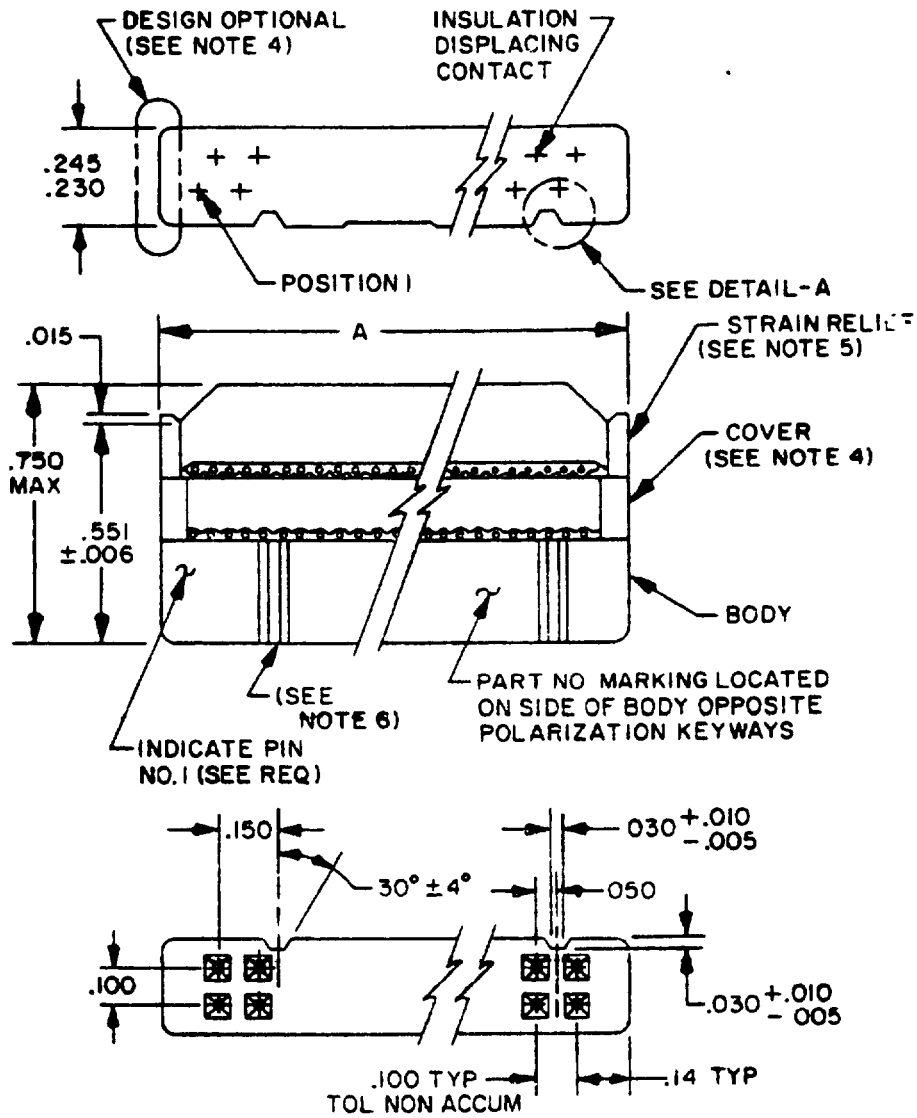
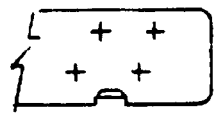
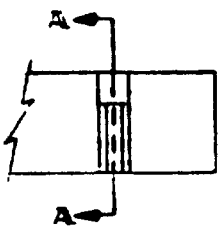


FIGURE 1. Dimensions and configurations.

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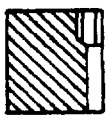
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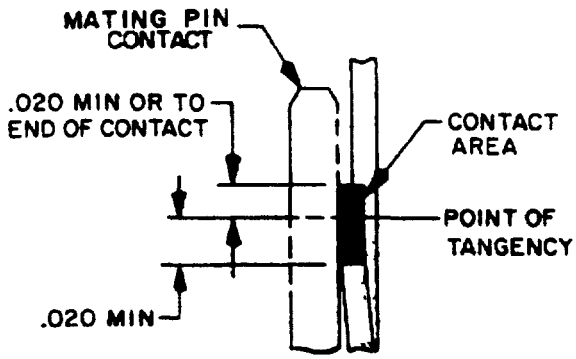


DETAIL-A

ALTERNATE POLARIZATION NOTCH
WITH NON FUNCTIONAL DETENT



SECTION A-A



FINISH LOCALIZATION AREA

Inches	MM
.005	.13
.006	.15
.010	.25
.015	.38
.020	.51
.030	.76
.050	1.27
.100	2.54
.14	3.6
.150	3.81
.230	5.84
.245	6.25
.551	14.00
.725	18.42

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25 mm.
3. Unless otherwise specified, tolerances are ±.005 (.13 mm) for three place decimals and ±.01 (.3 mm) for two place decimals.
4. Method of connection of cover to body optional but shall meet the requirements of MIL-C-83503 and this specification sheet.
5. Strain relief shall be provided. Strain relief design is optional (may be either separate or an integral part of cover).
6. The polarizing feature on this end is non-existent on 10 and 14 position connections.

FIGURE 1. Dimensions and configuration - Continued.

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TABLE I. Dash numbers and characteristics.

Dash number	Number of contacts	Dimension A inches 1/	Mating connector	
			Header (straight) M83503/21- or M83503/25-	Header (right angle) M83503/20- or M83503/24-
01	10	.68 (17.3)	01,02,03	01,02,03
02	14	.88 (22.4)	04,05,06	04,05,06
03	16	.98 (24.9)	07,08,09	07,08,09
04	20	1.18 (30.0)	10,11,12	10,11,12
05	24	1.38 (35.1)	---	---
06	26	1.48 (37.6)	13,14,15	13,14,15
07	30	1.68 (42.7)	---	---
08	34	1.88 (47.8)	16,17,18	16,17,18
09	40	2.18 (55.4)	19,20,21	19,20,21
10	50	2.68 (68.1)	22,23,24	22,23,24
11	60	3.18 (80.8)	25,26,27	25,26,27
12	64	3.38 (85.9)	28,29,30	28,29,30

1/ Metric equivalents are in parentheses.

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1 and table I.

Interface: Round conductor flat cable, .050 centers, to .100 x .100 inch spaced; .025 inch round or square posts, .230 minimum length. (NOTE: These connectors are not intended to be stacked.)

Temperature range: -55°C to +120°C.

Voltage rating. 300 V rms.

Current rating: 1 ampere.

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Materials:

Contacts. Beryllium copper per QQ-C-530 or QQ-C-533 phosphor bronze per QQ-B-750 with gold plating in the contact area

Housing, cover, and strain relief Glass reinforced polyester per MIL-M-24519. Cover retaining device may be made of QQ-S-763 class 301 stainless steel or other materials specified in MIL-C-83503.

Conductor accommodation 28 AWG stranded.

Test wire size: 28 AWG stranded.

Assembly tool: M22520/32.

Dielectric withstanding voltage (at sea level) 500 V rms, minimum
(at altitude) 200 V rms, minimum

Contact retention Not applicable.

Contact resistance. 50 milliohms, maximum.

Marking:

Part number: M83503/7-(dash number from table I).

Contact identification. Molded triangle "Δ" or notch to show number 1 contact, or raised numbers to show position number "1".

NOTE: Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodian:
Air Force - 85

Review activities:
Air Force - 11, 17, 99
DLA - ES

Agent:
DLA - ES

Preparing activity:
Air Force - 85

(Project 5935-F159-2)

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